

ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Sentiment in movie reviews

TF-IDF and SVM on IMDb

GROUP No.03
NGUYEN DOAN ANH HUY
NGUYEN PHUC HUNG
DAO ANH QUAN

ONE LOVE. ONE FUTURE.

A small word can change the whole review

Illustrative review snippets

“A good movie.”

Positive overall view

“Not a good movie.”

Negative overall view

Word order

Do bigrams help?

Kernel choice

Is RBF worth its cost?

Preprocessing

What should we remove?

One prediction for the review’s overall sentiment

Two runs answer different questions

Earlier CPU study

scikit-learn

20,000-review final training

5,000-review tuning subset

Earlier 6,000-review test sample

Main GPU study

cuML on two T4 GPUs

20,000-review training

Three-fold CV on the full train sample

New 6,000-review holdout

CPU evidence explains cost. GPU evidence supports the main findings.

A new holdout for a fairer evaluation

IMDb English movie reviews

20,000

Training reviews
10,000 positive and 10,000 negative

6,000

Holdout reviews
3,000 per class

Normalized text hashes and duplicate filtering
Earlier test IDs reconstructed and excluded

Historical ID reconstruction depends on the assumed source ordering.

How a review becomes a prediction

Review

"Not a good movie."

TF-IDF features

not good movie
not good good movie

SVM decision

Positive or negative

Training fold

Learns vocabulary and IDF

Validation fold

Uses the learned representation

Validation reviews never fit the vocabulary or IDF

Six representations and two kernels

Representation	Linear SVC	RBF SVC
Unigram	3 C settings	3 C × 3 gamma
Unigram + bigram	3 C settings	3 C × 3 gamma
Up to trigram	3 C settings	3 C × 3 gamma
Cleaned bigram	3 C settings	3 C × 3 gamma
Drop movie / film unigrams	3 C settings	3 C × 3 gamma
Drop containing n-grams	3 C settings	3 C × 3 gamma

216

CV fits

72 configurations
3 folds
12 final refits

The champion is chosen by CV before holdout evaluation

About nine in ten reviews classified correctly

89.30%

Holdout accuracy

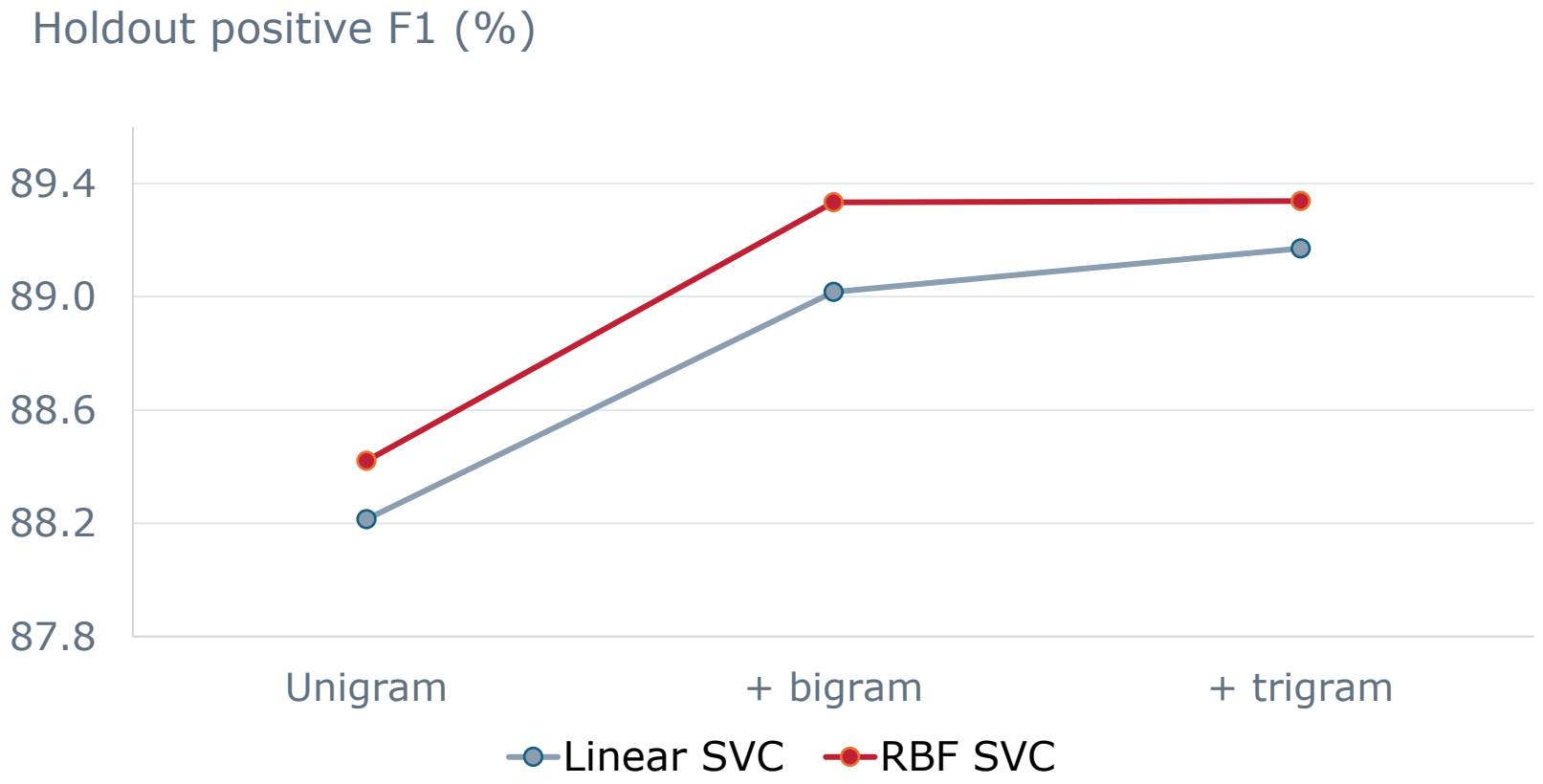
Positive F1	89.385%
Precision	88.681%
Recall	90.100%

Cleaned bigram RBF
C = 2, gamma = scale

True / predicted	Negative	Positive
Negative	2,655	345
Positive	297	2,703

5,358 correct
642 mistakes

Bigrams help more than trigrams



+0.913 pp

RBF bigram over unigram

+0.005 pp

Extra gain from trigram

All raw representations share a 12,000-feature budget

A higher score can still be an uncertain gain

Compared with raw bigram RBF

Configuration	F1 change	95% interval
Cleaning + bigram RBF	+0.052 pp	−0.254 to +0.349 pp
Trigram RBF	+0.005 pp	−0.321 to +0.330 pp
Bigram linear SVC	−0.316 pp	−0.669 to +0.046 pp
Unigram RBF	−0.913 pp	−1.525 to −0.338 pp

+0.316 pp

RBF's lead over linear bigram

95% interval
−0.046 to +0.669 pp

The RBF versus linear interval includes zero

Reported intervals condition on fitted models. Retraining variation is not included.

Should we remove movie and film?

“good movie”

Remove only the two unigrams

“bad film”

Or remove every containing n-gram

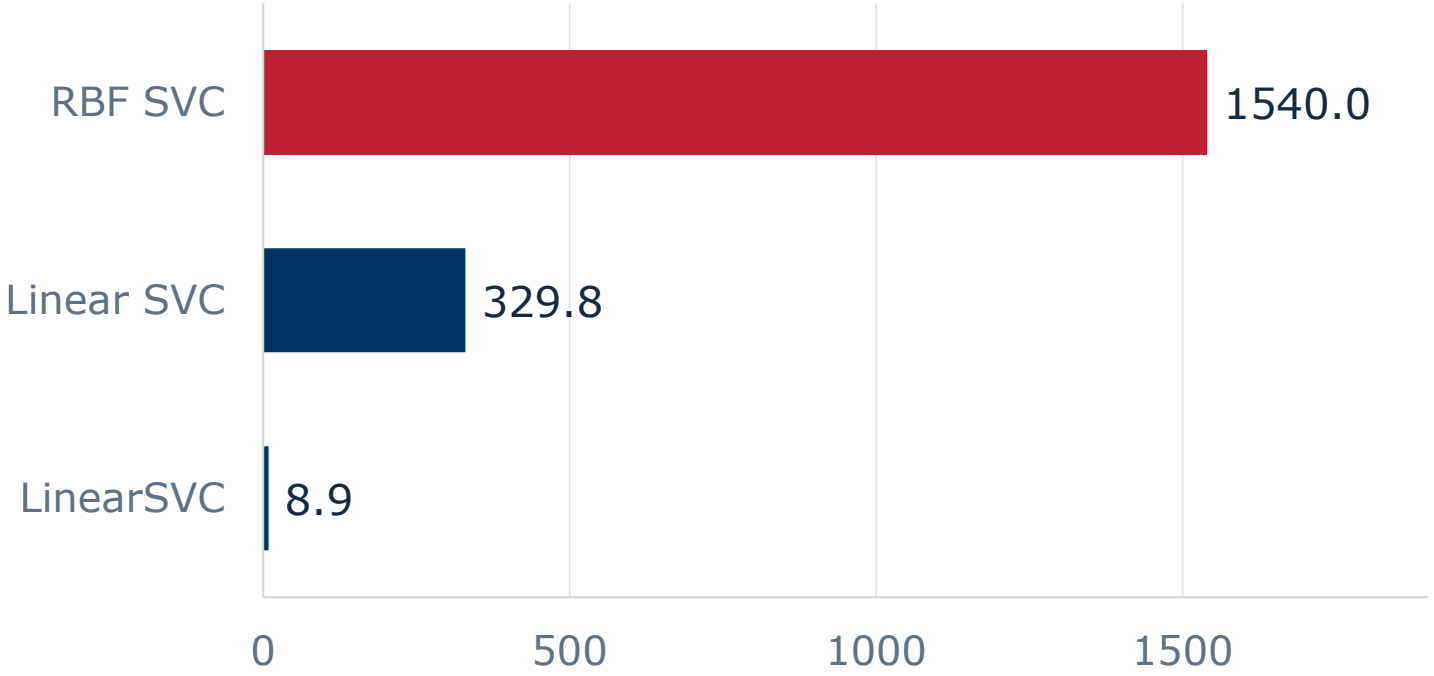
Feature removal	RBF F1 change	Linear F1 change
Two unigram columns	−0.018 pp	−0.055 pp
All containing n-grams	−0.187 pp	+0.142 pp

Neither removal shows a clear, consistent benefit

All four reported intervals include zero. Each pipeline retunes its parameters.

Almost the same accuracy, very different cost

Earlier CPU study: pipeline fit time in seconds



~173×

Longer RBF fit
than LinearSVC

Accuracy
LinearSVC 89.500%
RBF SVC 89.517%

LinearSVC is an efficient baseline in this CPU setting

What makes the results worth trusting?

216 fold results

Recalculation reproduces all 72 summaries and 12 selections.

Shuffled labels return to chance

49.925% linear

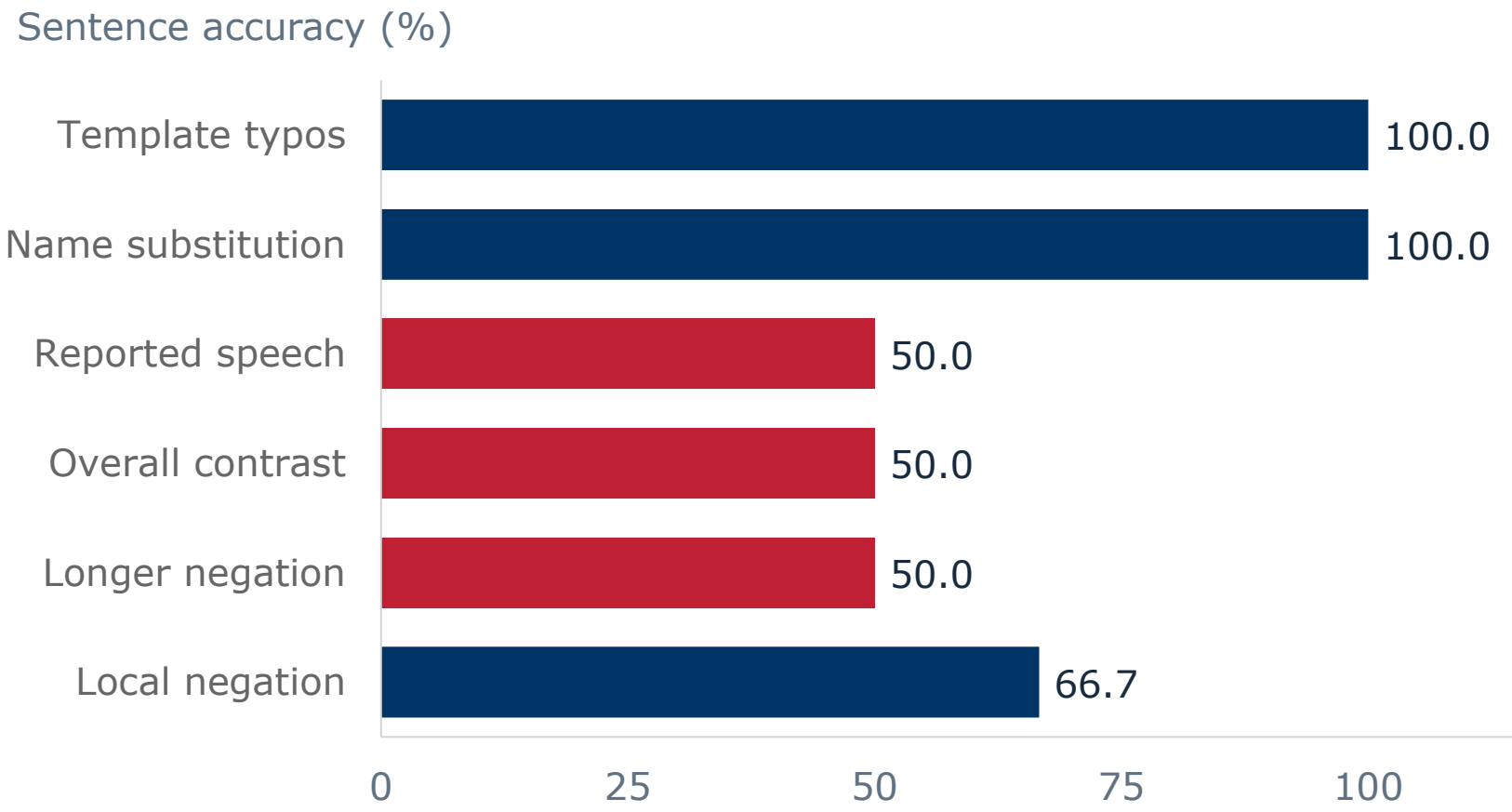
50.045% RBF

A gap still deserves attention

Training-fold F1 exceeds CV F1 by 10.45 points.

Additional hash and export checks are reported in RUN_REVIEW, not repeated from this small ZIP.

Strong IMDb scores do not cover every sentence



0 / 6

Pairs fully correct in each of the three red groups

12 sentences per group

These are small template tests. Manual label review remains unfinished.

What happens when the review changes its mind?

Clear praise

"A wonderful film with excellent acting and a moving story."

POSITIVE

Praise followed by an overall rejection

"The acting was excellent, but overall **I disliked the film."**

POSITIVE

Saved model prediction

CPU demo: notebook, CLI or Streamlit. Decision margin is not a probability.

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Thank you

Bigrams help

The clearest representation gain in this run.

Small differences remain uncertain

RBF and cleaning gains need qualification.

Language limits remain

Longer negation and discourse need more work.

Questions?